THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · MEDIUM LEVEL UNIT TEST SET 1

Chapters 11–14: **Exploring Some Geometric Themes, Tales by Dots and Lines, Algebra Play, Area** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A prism has 18 edges. The number of its faces is: 1

(a) 8 **(b)** 6 **(c)** 9 **(d)** 7

2. The mean mark of 10 students is 40 and of another 15 students is 60. The mean of all 25 students is: 1

(a) 50 **(b)** 54 **(c)** 1300 **(d)** 52

3. On expanding $3(x + 4) + 2x$ we get: 1

(a) $5x + 12$ **(b)** $3x + 14$ **(c)** $5x + 4$ **(d)** $6x + 12$

4. An L-shaped figure is made by removing a 8 cm by 6 cm rectangle from a 20 cm by 15 cm rectangle. Its area is: 1

(a) 300 cm² **(b)** 348 cm² **(c)** 108 cm² **(d)** 252 cm²

5. A pyramid has 7 vertices. The number of its edges is: 1

(a) 12 **(b)** 14 **(c)** 8 **(d)** 6

6. The mean of 59, 36, 46, 44, 50, 11 is 41. If 46 is replaced by 52, the new mean is: 1

(a) 41 **(b)** 49.67 **(c)** 42 **(d)** 47

7. $(x + 2)(x + 5)$ equals: 1

(a) $2x + 7$ **(b)** $x^2 + 7x + 10$ **(c)** $x^2 + 7$ **(d)** $x^2 + 10x + 7$

8. A parallelogram has base 18 cm and height 10 cm. If the base is changed to 15 cm (same area, other side), the corresponding height (in cm) is: 1

(a) 12 **(b)** 15 **(c)** 13 **(d)** 165

9. All 8 faces of a polyhedron are triangles. The number of its edges is: 1

(a) 16 **(b)** 12 **(c)** 8 **(d)** 24

10. The median of 46, 96, 29, 15, 31, 39, 70, 52 is: 1

(a) 40 **(b)** 42.5 **(c)** 31 **(d)** 47.25

11. If $a = 9$ and $b = 4$, the value of $3a - 5b + 8$ is: 1

(a) 15 **(b)** 30 **(c)** 14 **(d)** 16

12. A rhombus has side 15 cm and one diagonal 18 cm. Its area is: 1

(a) 108 cm² **(b)** 216 cm² **(c)** 270 cm² **(d)** 60 cm²

13. A polyhedron has 12 faces and 30 edges. Its number of vertices is: 1

(a) 20 **(b)** 44 **(c)** 18 **(d)** 40

14. The mean of 5 numbers is 28. The number to be added so that the mean becomes 30 is: 1

(a) 30 **(b)** 45 **(c)** 40 **(d)** 2

15. A rectangle has length $(3x + 2)$ cm and breadth $(2x + 5)$ cm. Its perimeter (cm) is: 1

(a) $10x + 7$ **(b)** $5x + 7$ **(c)** $10x + 14$ **(d)** $6x^2 + 10$

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. A cuboid is 15 cm long, 9 cm wide and 6 cm high. Find the sum of lengths of all its edges and the number of edges. 2

17. The mean marks of 15 boys is 60 and of 10 girls is 70. Find the mean mark of the whole class of 25. 2

18. Expand and simplify $2(x + 5) + 3(x - 1)$. 2

19. A rectangle has perimeter 80 cm and length 26 cm. Find its breadth and area. 2

20. A prism has 36 edges. Find its number of faces and vertices. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. A cube of side 4 cm is painted all over and cut into 1 cm cubes. Find the number of cubes with 3

(i) exactly three painted faces
(ii) no painted face
(iii) at least one painted face.

22. The mean weight of 10 students is 42 kg and of another 15 students is 48 kg. Find the mean weight of all of them. 3

23. The sides of a triangle are $(2x + 3)$ cm, $(3x + 1)$ cm and $(4x + 5)$ cm. Find its perimeter, and its value when $x = 2$. 3

24. A rectangular park is 40 m by 30 m. A path 2 m wide runs inside along its boundary. Find the area of the path. 3

OR

All 10 faces of a polyhedron are triangles. Find the number of edges and vertices. 3

25. The mean of 5 numbers is 26. If one number 49 is replaced by 59, find the new mean and the change in the mean. 3

OR

If $x + y = 11$ and $xy = 28$, find $x^2 + y^2$. 3

MASTERMIND

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. A prism and a pyramid both have a pentagonal base. Find the difference between the number of — 5

(i) faces

(ii) vertices

(iii) edges of the prism and the pyramid.

OR

The temperatures (°C) at 8 am, 10 am, 12 noon, 2 pm, 4 pm and 6 pm were 18, 22, 27, 30, 26, 21. — 5

(i) Find the rise or fall between each consecutive pair.

(ii) Find the mean temperature.

(iii) In which interval was the change largest?

27. Expand and simplify $(x+3)(x+5) - x(x+4)$. Find its value at $x = 2$. — 5

OR

A figure is made of a rectangle 20 cm by 10 cm with a right triangle of base 20 cm and height 8 cm placed on one of its longer sides. Find the total area and the area of the triangle as a percentage of the total. — 5

END OF QUESTION PAPER — Total: 50 Marks

MASTERMIND